

Tailwind CSS Classes and Their Uses

Layout

- container: Centers the content and applies max-width based on screen size.
- block: Displays an element as a block-level element.
- inline-block: Displays an element as an inline block-level element.
- flex: Creates a flex container for flexible layouts.
- grid: Creates a grid container for grid-based layouts.
- hidden: Hides the element.

Flexbox and Grid

- flex-row: Sets the flex items to align in a row.
- flex-col: Sets the flex items to align in a column.
- justify-center: Centers flex items along the main axis.
- items-center: Centers flex items along the cross axis.
- grid-cols-3: Defines a grid with 3 columns.
- gap-4: Adds spacing between grid or flex items.

Spacing

- p-4: Applies padding of 1rem (16px) on all sides.
- m-4: Applies margin of 1rem (16px) on all sides.
- px-2: Applies horizontal padding of 0.5rem (8px).
- py-2: Applies vertical padding of 0.5rem (8px).
- space-x-4: Adds horizontal spacing between inline children.

Typography

- text-lg: Applies a font size of 1.125rem (18px).
- font-bold: Makes the text bold.
- text-center: Centers the text horizontally.
- text-gray-500: Applies a gray color to the text.
- uppercase: Transforms the text to uppercase.

Background

- bg-blue-500: Applies a blue background color.
- bg-gradient-to-r: Creates a gradient background from left to right.
- from-green-500: Defines the start color for a gradient.
- bg-cover: Sets the background image to cover the element completely.

Borders

- border: Adds a border of 1px solid.
- rounded: Applies a small border radius to the element.
- rounded-full: Makes the element fully rounded, like a circle.
- border-gray-500: Applies a gray color to the border.

Shadows and Effects

- shadow: Adds a small box shadow.
- shadow-lg: Adds a large box shadow.
- opacity-75: Sets the opacity to 75%.

Interactivity

- cursor-pointer: Changes the cursor to a pointer on hover.
- hover:bg-blue-500: Changes the background color to blue on hover.
- focus:outline-none: Removes the focus outline.

Responsive Design

- sm:w-full: Sets the width to 100% on small screens.
- md:p-4: Applies padding of 1rem (16px) on medium screens.
- lg:text-lg: Sets the font size to large on large screens.

Layout

- **Container:** container
- **Display:** block, inline-block, flex, inline-flex, grid, inline-grid, hidden
- **Position:** absolute, relative, fixed, sticky
- **Top/Right/Bottom/Left:** top-0, right-0, bottom-0, left-0, inset-0, inset-x-0, inset-y-0
- **Z-Index:** z-0, z-10, z-20, etc.
- **Overflow:** overflow-hidden, overflow-auto, overflow-scroll
- **Object Fit:** object-contain, object-cover, object-fill

Flexbox and Grid

- **Flexbox Container:** flex, inline-flex
- **Flexbox Properties:**
 - o flex-row, flex-col, flex-wrap, flex-nowrap
 - o justify-start, justify-center, justify-end, justify-between, justify-around
 - o items-start, items-center, items-end, items-baseline
 - o self-start, self-center, self-end
- **Grid Container:** grid, inline-grid
- **Grid Properties:**
 - o grid-cols-1, grid-cols-2, grid-rows-1, grid-rows-2

- o gap-1, gap-2, gap-x-1, gap-y-1

Spacing

- **Padding:** p-1, p-2, px-1, py-1, pt-1, pr-1, pb-1, pl-1
- **Margin:** m-1, m-2, mx-1, my-1, mt-1, mr-1, mb-1, ml-1
- **Space Between:** space-x-1, space-y-1

Sizing

- **Width:** w-auto, w-full, w-1/2, w-1/4, max-w-lg
- **Height:** h-auto, h-full, h-screen, h-1/2
- **Min/Max Width/Height:** min-w-0, max-w-full, min-h-0, max-h-full

Typography

- **Font Size:** text-xs, text-sm, text-base, text-lg, text-xl, text-2xl
- **Font Weight:** font-thin, font-light, font-normal, font-bold, font-extrabold
- **Text Align:** text-left, text-center, text-right, text-justify
- **Text Color:** text-black, text-white, text-gray-500, text-red-500
- **Text Decoration:** underline, line-through, no-underline
- **Text Transform:** uppercase, lowercase, capitalize
- **Leading (Line Height):** leading-none, leading-tight, leading-loose

Background

- **Color:** bg-black, bg-white, bg-gray-500, bg-red-500
- **Gradient:** bg-gradient-to-r, bg-gradient-to-l, from-blue-500, to-green-500
- **Position:** bg-center, bg-top, bg-bottom, bg-left

- **Size:** bg-cover, bg-contain, bg-auto

Borders

- **Border Width:** border, border-0, border-2, border-4
- **Border Radius:** rounded, rounded-full, rounded-lg, rounded-none
- **Border Color:** border-black, border-white, border-gray-500
- **Divide Borders:** divide-x, divide-y, divide-gray-500

Shadows and Effects

- **Box Shadow:** shadow-sm, shadow, shadow-lg, shadow-none
- **Opacity:** opacity-0, opacity-25, opacity-50, opacity-75, opacity-100
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Interactivity

- **Cursor:** cursor-pointer, cursor-not-allowed
- **User Select:** select-none, select-text, select-all
- **Hover, Focus, Active:** hover:bg-blue-500, focus:outline-none, active:bg-red-500

Responsive

- Prefixes:
 - o sm:, md:, lg:, xl:, 2xl:
 - o Example: sm:w-full, md:p-4, lg:text-lg